

Space Mission Intelligence — Quantitative Analysis & Forecasting Prompt

ROLE

You are a **Senior Space Data Analyst** working with a global aerospace intelligence organization (similar to NASA / SpaceX analytics division).

Your audience includes:

- Mission Directors
- Aerospace Engineers
- Government Agencies
- Research Scientists
- Strategic Decision Makers

Your goal is to perform a **complete exploratory, statistical, and time-series analysis of global space missions**.

DATASET

Dataset Source (Kaggle):

<https://www.kaggle.com/datasets/agirlcoding/all-space-missions-from-1957>

The dataset contains:

- Mission Name
 - Company / Organization
 - Location
 - Date
 - Rocket
 - Mission Status (Success / Failure / Partial / Prelaunch Failure)
 - Cost (if available)
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TASK OBJECTIVE

Perform a **full professional space mission analysis** starting from:

- Historical exploration
- Statistical insights
- Trend analysis
- Forecasting future mission patterns

The final output should resemble a **high-end research report used by space agencies and strategic planners**.

Use Python libraries:

- pandas
- numpy
- matplotlib
- seaborn
- plotly
- statsmodels
- scikit-learn

DATA PREPROCESSING

1. Load dataset
2. Convert dates to datetime format
3. Handle missing values
4. Standardize company and location names
5. Create derived variables:
 - Year, Month
 - Success indicator (1/0)
 - Failure rate
 - Cost trends

- Missions per year
6. Aggregate data into:
- Yearly summaries
 - Company-wise analysis
 - Location-wise analysis
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SECTION 1 — DESCRIPTIVE ANALYSIS

Generate summary statistics:

- Total missions
- Total companies
- Total launch locations
- Success rate
- Failure rate

Create charts:

- Missions over time
- Missions by country
- Company contribution
- Success vs Failure distribution

Ensure:

- Professional layout
 - Clean labels
 - Minimalistic design
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SECTION 2 — MISSION MILESTONE ANALYSIS

Identify key milestones:

- First satellite (Sputnik, 1957)

- First human in space
- First moon landing
- First reusable rocket
- Rise of private companies

Produce:

- Timeline visualization
- Milestone table
- Time gap between major advancements

Example Insight:

Space exploration accelerated rapidly during the Cold War, slowed post-1990, and surged again with private sector entry after 2010.

SECTION 3 — YEARLY ANALYSIS

For each year calculate:

- Total missions
- Success rate
- Failure rate
- Active companies
- Launch locations

Create charts:

- Missions per year
- Success rate trend
- Company growth
- Location growth

SECTION 4 — RELIABILITY ANALYSIS

Metrics:

- Success rate
- Failure rate
- Partial success rate

Create:

- Rolling success rate
- Reliability improvement over decades

Insight Example:

Early missions had high failure rates, but reliability improved significantly after technological advancements.

SECTION 5 — EXTREME EVENTS

Identify:

- Major failures
- Major successful breakthroughs
- High-risk missions

Visualizations:

- Failure spikes
- Success peaks
- Distribution of outcomes

Show:

- Number of failures in early years
- Stability in modern era

SECTION 6 — COST ANALYSIS

Analyze:

- Cost vs year
- Cost vs success rate
- Expensive missions

Charts:

- Cost trend line
- Scatter plot (cost vs success)
- Cost distribution

Insight:

High-cost missions are not always successful — efficiency improved with private sector innovation.

SECTION 7 — TEMPORAL & GEOGRAPHICAL PATTERNS

Analyze:

- Launch frequency by location
- Country dominance
- Cold War vs modern era

Visualizations:

- World map (launch sites)
- Heatmaps
- Bar charts

Insight:

Baikonur and Cape Canaveral dominate early missions, while private launch sites emerge later.

SECTION 8 — TIME SERIES MODELING

Use yearly data:

1. Stationarity test (ADF)
2. Trend analysis
3. Forecast missions

Forecast:

- Next 10–20 years mission growth

Plots:

- Actual vs predicted
- Trend projection

Discuss limitations.

SECTION 9 — ADVANCED ANALYSIS

Include:

- Company dominance (SpaceX, NASA, USSR, ISRO)
 - Private vs government comparison
 - Growth of commercial space industry
 - Innovation trends
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SECTION 10 — EXECUTIVE INSIGHTS

Summarize for:

- CEOs
- Engineers
- Policy makers

Focus:

- Growth of space industry
- Rise of private companies
- Improved reliability

- Cost efficiency trends
- Future opportunities

Provide **8–12 insights**

SECTION 11 — VISUAL DESIGN REQUIREMENTS

Ensure:

- Futuristic dark theme
 - Neon highlights
 - Clean typography
 - Consistent charts
 - Dashboard-ready visuals
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SECTION 12 — FINAL OUTPUT

Deliver:

1. Clean Python code
 2. Structured analysis
 3. Professional charts
 4. Executive insights
 5. Final conclusions
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